WASP-11b

R-1637

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-10_130918` · created 2026-08-02T06:28:25+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456553.92118 +/- 0.00075 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.114 +/- 0.051
Transit depth (Rp/Rs)^2	1.3 +/- 1.15 [%]
Orbital Inclination (inc)	86.31 +/- 1.66
Airmass coefficient 1 (a1)	1.03 +/- 0.32
Airmass coefficient 2 (a2)	0.19 +/- 1.01
Scatter in the residuals of the lightcurve fit is	30.95 %
Best Comparison Star	#2 - [85, 241]
Optimal Aperture	24.82
Optimal Annulus	99.29
Transit Duration (day)	0.065 +/- 0.035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.114 ± 0.051 vs 0.131 (deviation 0.33σ)
Duration fit vs published	0.065 d vs 0.1075 d (deviation 1.21σ)
Mid-time O–C	-1.7 min
Depth SNR	2.2
Residual scatter	30.95 %

Light curve

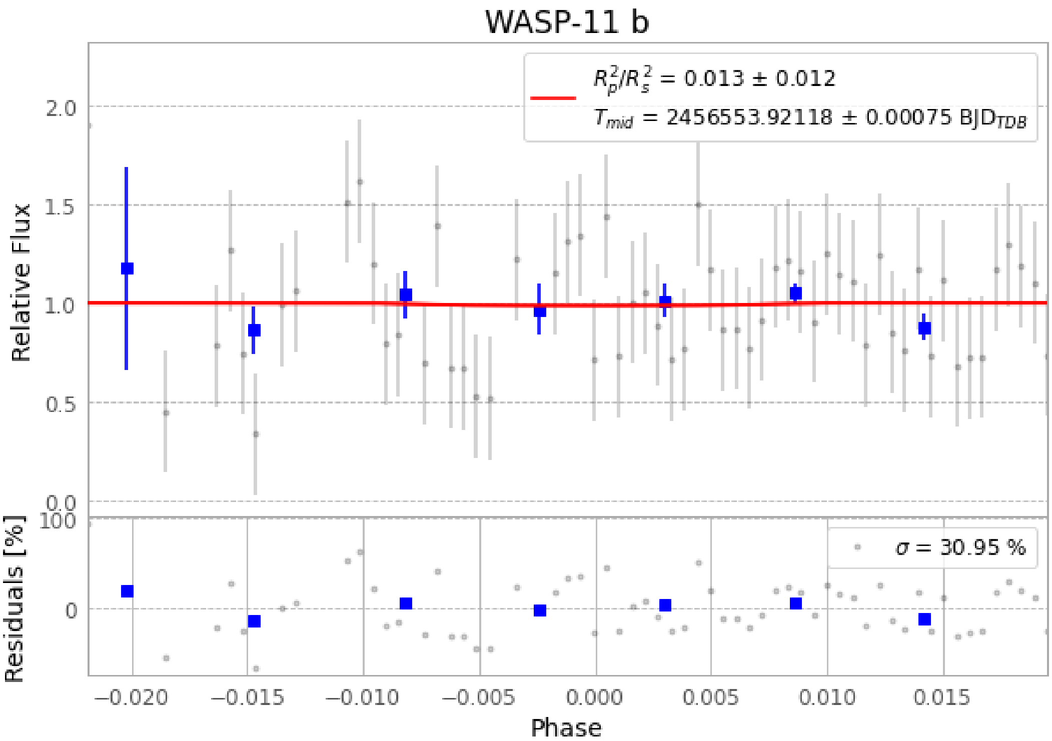

Figure 1 — FinalLightCurve_WASP-11 b_2013-09-18.png (SHA-256 d677ac5e6174ef33…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [26, 143], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 bd6934c8d28d8bcd…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

75 FITS frames (49 MB), `HATP-10130918080312.FITS` → `HATP-10130918114512.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-130918.log (6dfb5d21e674…), FinalLightCurve_WASP-11 b_2013-09-18.png (d677ac5e6174…), FinalLightCurve_WASP-11 b_2013-09-18.csv (a2812664af20…)

Compute resources

Wall time 1011.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-08-02 06:28Z.